

Offline: train once and compile several static engines

Shared supernet
one training process

extract capacity

Four static capacities
tiny | small | medium | large

compile

compile

TensorRT engines
static FP16 graphs

Hailo-8 HEFs
static dataflow graphs

benchmark

benchmark

Measured route costs
backend-specific latency table

measured costs

Runtime: measure visual risk and execute one feasible engine

Adverse-condition image
fog | night | rain | snow

input

Tiny probe inference
cheap segmentation pass

logits

Visual-risk score
mean pixel entropy

shared score

Candidate-specific error
four calibrated estimates

expected errors

Budget-constrained selector
risk target + latency budget

choose engine

Selected static subnet
one compiled engine

execute

Segmentation output
19-class prediction

hard budget

Latency budget
deployment constraint